SpotCast

Synopsis for the app project

SWMAD-01 Mobile Application Development

Group members and study numbers: [FILL IN: name, study number, one line per member]

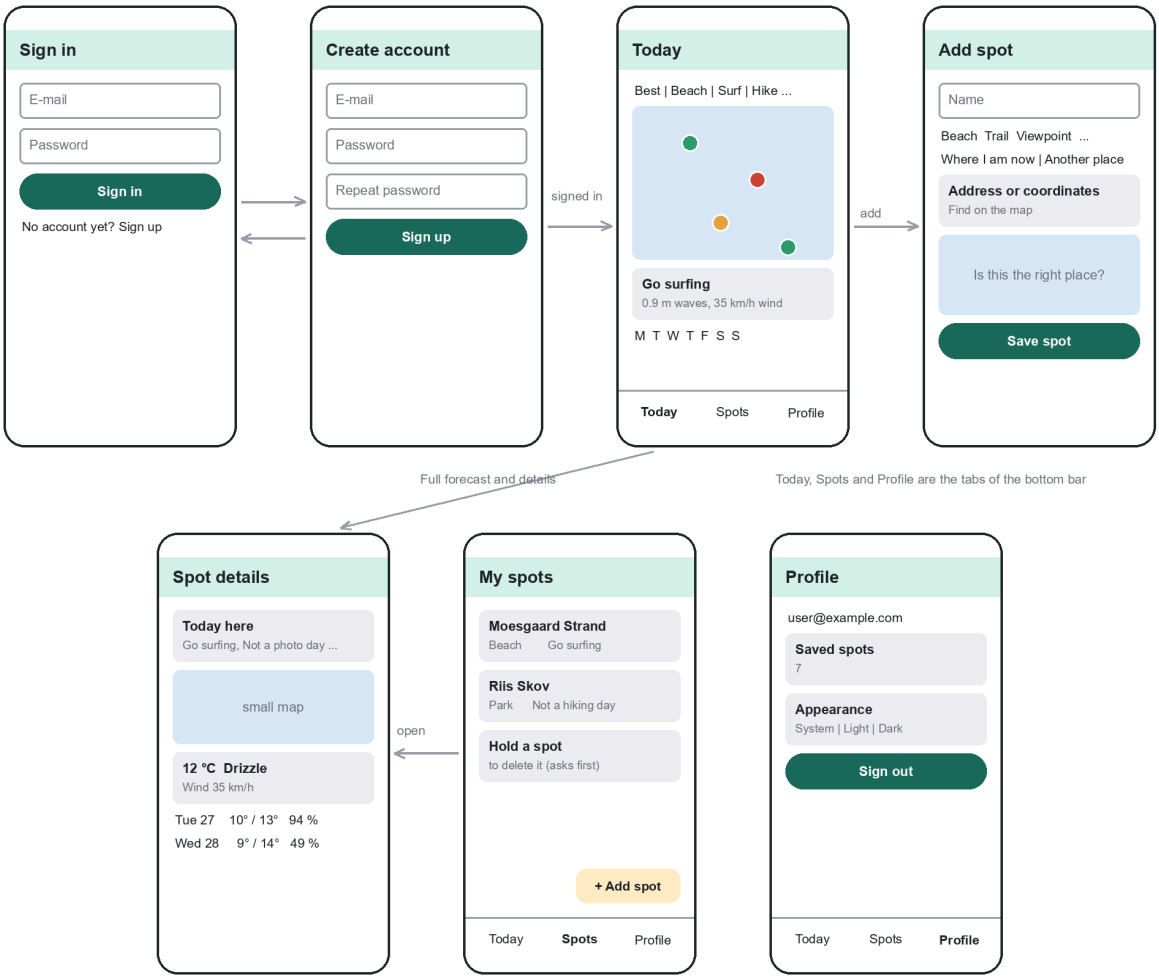
App vision

People who spend time outdoors keep a mental list of places: a beach that gets waves, a forest for a walk, a pier with a good sunset. A weather app gives them numbers for the nearest town, but not the answer they need: is today a good day to do this, here?

SpotCast answers Go, Maybe or Skip for every saved spot and five activities: a beach day, surfing, hiking, photos and a picnic. The app opens on a map where each pin has the colour of today's answer. Tapping a pin explains the answer, suggests a better activity for that spot, and shows the best day of the week. Spots are saved where the user stands, or anywhere by address, and stay in the user's account.

Screens

The app has five screens besides sign in and sign up. The layout below shows all seven.



- **Today**: the map with a pin per spot coloured by verdict, an activity bar, and a card with the full report for the selected spot.
- **Spots**: the list of spots with today's best activity; holding a spot deletes it after a confirmation.
- **Add spot**: name, category and the position, from GPS or from an address or coordinates that the user confirms on a map.
- **Spot details**: the verdict for all five activities, a small map, the weather now and the next

seven days.

- **Profile**: the account, the spots per category, the light, dark or system theme, and sign out.

User stories

- As someone with a free afternoon, I want to see which of my spots are worth it today, so that I do not have to read seven forecasts.
- As a photographer who also surfs, I want to pick an activity and be told when another one suits a spot better.
- As a user, I want to know why a spot is a Skip and which day this week is better.
- As a hiker, I want to save the place where I stand with one button, or a place I have not been to by its address.
- As a user, I want to delete a spot without risking deleting one by accident.
- As a user with a phone and a tablet, I want my spots to follow my account.

Technologies

- Kotlin, coroutines and Jetpack Compose with Compose Navigation.
- Firebase Authentication with e-mail and password (mandatory).
- Firestore database for the spots of each user, with real-time updates (mandatory).
- Ktor HTTP client calling the Open-Meteo forecast and marine APIs, which are not Firebase services and are called without an SDK (mandatory), and Nominatim for address search.
- GPS through the fused location provider, and a vector map with MapLibre and the free OpenFreeMap tiles (the additional technology).